مدارس الريادة العالمية
AL-REEYADA INTERNATIONAL SCHOOL

AL-REEYADA INTERNATIONAL SCHOOL

MAHASEN ARAMCO, AL-AHSA, KINGDOM OF SAUDI ARABIA

SCIENCE 6 WORKSHEET NO. 3

LESSON 3: THE STATE OF MATTER

Reminder: Please print a copy of this worksheet and bring it on NOVEMBER 4-5, 2026 (Wednesday-Thursday), WITHOUT ANSWER.

NAME: _____

Roll No. _____ Grade/Section: 6 _____ Date: _____

-PAPER 2-

I. MULTIPLE CHOICE: Choose the most appropriate answer and write the corresponding letter in the blank.

_____1. What was Otto von Guericke's main scientific interest?

A. The existence of a vacuum B. Inventing the steam engine

C. The invention of the microscope D. Discovering the law of gravity

_____2. According to the particle model, what happens to the particles in a liquid?

A. They vibrate in a fixed position.

B. They are very far apart and move at high speed.

C. They can change position by sliding over each other.

D. They are held by strong forces in an orderly arrangement.

_____3. What happens to sound in a vacuum?

A. It becomes louder. B. It echoes. C. It cannot be heard. D. It travels faster.

_____4. The object used to seal the flask during the chemical reaction.

A. Stopper B. Thistle funnel C. Glass trough D. All of them

-PAPER 4-

I. Fill in the blanks.

1. When a ball is thrown in space, theoretically it would travel in a _____ line.

2. Unlike gases, solids and liquids particles are hard to _____ because their particles are packed closely together.

3. Otto von Guericke proved that a _____ couldn't be heard in a vacuum.

4. A solid chemical is mixed with a liquid chemical to produce a _____ during the chemical reaction.

5. Space is a vacuum, as there is no _____ to slow objects down.

6. The volume of a gas can _____ depending on the size of the container it occupies.

II. Match the following statements of column B with A.

S. No.	Column A	Answer	Column B
1	Carbon dioxide		(a) Have definite shape and volume
2	Solids		(b) It has particles spread over great distances from each other.
3	Thistle funnel		(b) It changes limewater from clear to cloudy.
4	Collide and move in any direction		(c) Apparatus used to pour down liquid chemical
5	Partial vacuum		(d) Particles of gases

PAPER 4

III. Define the following terms.
A) Forces of attraction

Ans: _____

_____

B) Vacuum

Ans: _____

_____

IV. Answer the following questions.

1. How do heat and light travel out into the space?

Ans: _____

_____

_____

V-A. Sequencing: Arrange the following statements in the correct order.

A. Tried different vessels, but they collapsed.
B. Otto von Guericke wanted to test whether a vacuum could really exist.
C. Placed a ringing bell inside the barrel and removed air.
D. He made an air pump to test his ideas.
E. He proved that sound couldn't be heard.
F. Used hollow copper hemispheres to make a globe.

Ans: _____, _____, _____, _____, _____, and _____

V-B. Sequencing: Arrange the following in order on how to test oxygen gas.

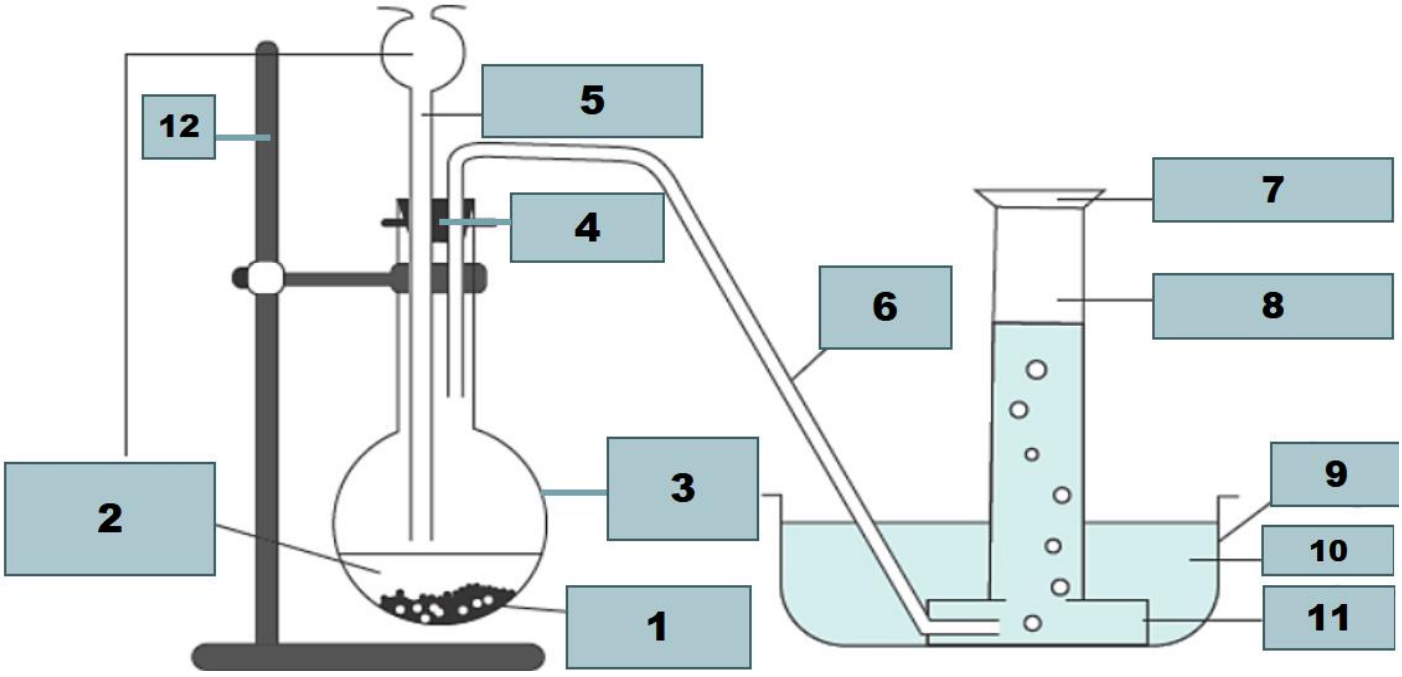
A. Show the splint towards the test tube containing oxygen.
B. The fire lights back (the splint produces a flame).
C. Pick up a splint and light it one end.
D. Blow out the flame on the splint but make sure it is still glowing.

Answer: _____, _____, _____, and _____

PAPER 6

I. Label the given experimental set-up and answer the following questions.

a. Write the answer in the space provided.



1. _____

2. _____

3. _____

4. _____

5._____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

b. Write the reaction taking place during the process.

_____ + _____ → _____ + _____

II. Read the statement clearly and answer the following questions.
A student was given two test tubes containing samples of two different gases. The student was asked to identify the gases provided. After the practical procedure, the observations were noted down as given in the table below. Complete the missing columns.

S. No.	Test	Result	Identified gas
1.	Took a splint, lighted at one end and blew the flame to get a glowing splint.		
2.		The solution changes from clear to cloudy	
3.	Took a splint and lighted at one end to get a burning a splint.		

Therefore, the gases present in the test tubes were:

Ans: _____

III. Draw a neat diagram of particle model of states of matter and complete the following table:

S. No.	State of matter	Diagram of particle model	Description of the particles
1.	Solids		**Arrangement:** _____ _____ **Separation:** _____ _____ **Motion:** _____ _____

2.	**Liquids**		**Arrangement:** _____ _____ **Separation:** _____ _____ **Motion:** _____ _____
3.	**Gas**		**Arrangement:** _____ _____ **Separation:** _____ _____ **Motion:** _____ _____